(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

VI Semester, March 2025

Paper Code: DA-304T **Sub.: Statistics, Statistical Modelling & Data Analytics**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No. 1

i. What are discrete and continuous variables? 2 (CO1)

ii. Give the relationship between mean, median and mode for positively skewed, negatively skewed and symmetric distributions. 2 (CO1)

iii. Give mathematical representations for simple and multi-linear regression models. 2 (CO2)

iv. Write properties of linear estimators. 2 (CO2)

v. State the assumptions made for unobservable errors in the regression models of Q. 1 (iii). 2 (CO2)

Q. No. 2

Explain in detail sampling distribution of the means. What is the significance of Central limit theorem in ascertaining the nature of such distributions? Hence, discuss sampling statistics for (a) sampling with and without replacement, and (b) large and small sampling cases. 10 (CO1)

Q. No. 3

i. (a) Find the pdf for the standard normal distribution. 2 (CO1)
(b) Explain the significance of chi-square distribution in statistics, and F-distribution in model (variable) selection. 3 (CO1, CO2)

ii. Two distances are measured as 27.3 centimeters (cm) and 15.6 cm with standard deviations (standard errors) of 0.16 cm and 0.08 cm, respectively. Determine the mean and standard deviation of (a) the sum and (b) the difference of the distances. 5 (CO1)

Q. No. 4

i. Explain least-squares method for estimation of regression parameters, and derive expressions for linear estimators. 5 (CO2)

ii. Write short notes on *any two* - 2x2.5 =5 (CO2)
(a) Gauss-Markov Theorem (b) Factorial Experiments
(c) Logistic regression model (d) Transformations & Box-Cox model

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Q. No. 4

i. Explain least-squares method for estimation of regression parameters, and derive expressions for linear estimators. 5 (CO2)

ii. Write short notes on *any two* - 2x2.5 =5 (CO2)
(a) Gauss-Markov Theorem (b) Factorial Experiments
(c) Logistic regression model (d) Transformations & Box-Cox model

(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] JUNE 2024

Paper Code: DA-304T ***Subject: Statistics, Statistical Modelling & Data Analytics***

Time: 3 Hours **Maximum Marks:75**

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit.

Q1 Attempt All **[3x5=15]**

a) State Gauss-Markov theorem in detail.

b) Define open set and closed set. Also give examples.

c) State Cayley- Hamilton theorem and why it is useful.

d) A variate X has the probability distribution

X	-3	6	9
P(X=x)	1/6	½	1/3

Find E(X) and $E(X^2)$. Hence evaluate Var(X).

e) Two bolts are drawn from a box containing 6 good and 8 defective bolts. Find the probability that the second bolt is good if the first one is found to be defective.

UNIT-I

Q2 a) A manufacturer knows that the condensers he makes, contain on an average 1% defective. He packs them in boxes of 100. What is the probability that a box selected at random will contain 3 or more defective condensers? **(5)**

b) Calculate the mean and standard deviation for the following: **(5)**

Size of item	6	7	8	9	10	11	12
Frequency	3	6	9	13	8	5	4

c) A sample of 100 iron bars is said to be drawn from a large number of bars whose lengths are normally distributed with mean 4 feet and S.D. 0.6 feet? If the sample mean is 4.2 feet, can the sample be regarded as a truly random sample? **(5)**

Q3 a) Define chi-square test as goodness of fit. A random number table of 250 digits showed the following distribution of digits 0,1,2,3,4,5,6,7,8,9.

Digit	0	1	2	3	4	5	6	7	8	9
Observed Frequency	17	31	29	18	14	20	35	30	20	36
Expected Frequency	25	25	25	25	25	25	25	25	25	25

Does the observed distribution differ significantly from expected distribution using a significance level of 0.01? Given that $\chi^2_{0.99}$ for a degree of freedom is 21.7. **(7.5)**

b) X is a normal variate with mean 30 and standard deviation is 5. Find the probability that: i) $26 \le X \le 40$ ii) $X \ge 45$ iii) $|X-30|>5$.
(Given that $P(0 \le z \le 1) = 0.3413$, $P(0 \le z \le 2) = 0.4772$, $P(0 \le z \le 0.8) = 0.2881$, $P(0 \le z \le 3) = 0.4986$). **(7.5)**

P.T.O.

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MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

6th Semester, April, 2024

Paper Code: DA-304T **Subject:** Statistics, Statistical Modelling and Data Analytics(SSMD)

Time:1 ½ Hrs. **Max.Marks: 30**

Note: Attempt Q.No. 1 which is compulsory and any two more questions from remaining.

Q.No.		Question	Max.Marks	CO(s)
1	a)	Three students A, B, and C are running in a race. A and B have the same probability of winning and each is twice as likely to win as C. Find the probability that B or C wins	3M	CO1
	b)	Define Properties of the Normal Distribution and also Define any two hypothesis tests	2M	CO1
	c)	Obtain the line of regression of y on x for the data given below:	3.5M	CO2

X	1.53	1.78	2.60	2.95	3.42
Y	33.50	36.30	40.00	45.80	53.50

	d)	Define Data Visualization and Least square method with a real life application						1.5M	CO2
2	a)	Find the mean and variance of uniform probability distribution $f(x)=\frac{1}{n}$ for x=1,2,...n						5M	CO1
	b)	For discrete probability distribution						5M	CO1

x	0	1	2	3	4	5	6	7
f	0	k	2k	2k	3k	K^2	$2k^2$	$7k^2+k$

Determine: (i)k (ii) Mean, (iii) Variance (iv) smallest value of x such that $P(X \le x) > \frac{1}{2}$

3	a)	State and prove Gauss Markov theorem	8M	CO2
	b)	Let V be a vector space in $\mathbb{R}^3$. Examine whether the following are the subspaces or not? (i) $W=\{(a,b,c) \mid a^2+b^2+c^2 \le 1\}$ and (ii) $W=\{(a,b,c) \mid a=b=c\}$	2M	CO2
4	a)	Fit a straight line to the following data considering y as the dependent variable:	8M	CO2

x	1	2	3	4	5
y	5	7	9	10	11

	b)	In a sample of 1000 people, 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice eater and wheat eater are equally popular at 1% level of significance	2M	CO1

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